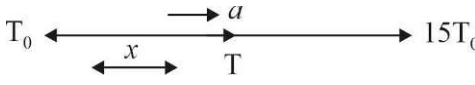


Solutions to JEE Advanced -10 | JEE 2022 | Paper-1

PHYSICS

1.(D) $a = \frac{15T_0 - T_0}{m_0} = \frac{14T_0}{m_0}$



$$T - T_0 = \frac{m_0}{L} x a = \frac{m_0}{L} \frac{14T_0}{m_0} x$$

$$\therefore T = T_0 + \frac{14T_0}{L} x = T_0 \left(1 + \frac{14x}{L} \right)$$

$$\frac{dx}{dt} = v = \sqrt{\frac{T}{\mu}}$$

$$\Rightarrow \frac{dx}{dt} = \sqrt{\frac{T_0 L}{m_0} \left(1 + \frac{14x}{L} \right)} \Rightarrow dt = \frac{dx}{\sqrt{\frac{T_0}{m_0} (L + 14x)}} \Rightarrow \int dt = \sqrt{\frac{m_0}{T_0}} \int_0^L \frac{dx}{\sqrt{L + 14x}}$$

$$\Rightarrow t = \frac{1}{7} \sqrt{\frac{m_0}{T_0}} \left[\sqrt{L + 14x} \right]_0^L = \sqrt{\frac{m_0}{49T_0}} (\sqrt{15L} - \sqrt{L}) \therefore t = \sqrt{\frac{m_0 L}{49T_0}} (\sqrt{15} - 1)$$

2.(BC) $H_1 = H_2 + H_3$

$$\Rightarrow KA \left(\frac{60 - T_M}{x} \right) = 2KA \left(\frac{T_M - T_N}{x} \right) + KA \left(\frac{T_M - T_N}{x} \right)$$

$$\Rightarrow 60 - T_M = 3T_M - 3T_N$$

$$\therefore 4T_M - 3T_N = 60 \quad \dots(i)$$

$$H_1 = H_4$$

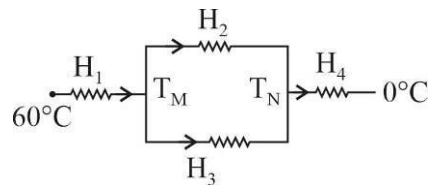
$$\Rightarrow KA \left(\frac{60 - T_M}{x} \right) = 2KA \left(\frac{T_N - 0}{x} \right)$$

$$\Rightarrow 60 - T_M = 2T_N$$

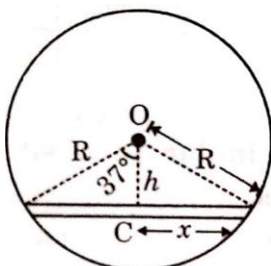
$$\therefore T_M + 2T_N = 60 \quad \dots(ii)$$

By (i) and (ii), $T_M = 27.3^\circ\text{C}$

$$T_N = 16.4^\circ\text{C}$$



3.(BD) Consider squirrel at distance x from midpoint C of platform.



F.B.D. of squirrel :

$$f = ma \quad \dots(i)$$

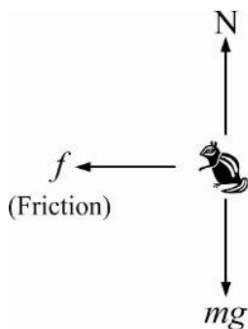
$$N = mg \quad \dots(ii)$$

F.B.D. of platform + Wheel cage

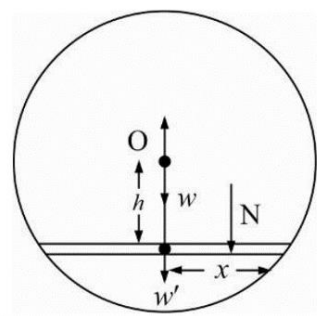
Platform + wheel cage remain at equilibrium only if $\tau_{pivot} = 0$, which is possible according to the following

F.B.D. only

$$\Rightarrow fh = Nx \Rightarrow mah = mgx \quad [\text{From equations (i) and (ii)}]$$



$$a = \frac{gx}{h} \text{ towards C (restoring acceleration)}$$



$\therefore$ Squirrel performs SHM about mean position C, angular frequency $\omega = \sqrt{\frac{g}{h}}$ and amplitude

$$A = \frac{3R}{5} \quad (\text{Half the platform's length})$$

In SHM, velocity at extreme should be zero.

Squirrel is at rest at one end of the platform at $t = 0$.

$$\text{Maximum velocity} = A\omega = \frac{3R}{5} \sqrt{\frac{g}{h}}$$

$$v_{\max} = \frac{3R}{5} \sqrt{\frac{g}{\frac{4R}{5}}} \Rightarrow v_{\max} = \sqrt{\frac{9Rg}{25} \cdot \frac{5}{4}} \Rightarrow v_{\max} = \sqrt{\frac{9Rg}{20}}$$

$$\text{Maximum acceleration} = \omega^2 A$$

$$a_{\max} = \frac{gA}{h} = \frac{g \frac{3R}{5}}{\frac{4R}{5}} = \frac{3g}{4}$$

- 4.(AD) Potential difference across R_1 increases $\therefore i_{R_1} \uparrow$
 Potential difference across R_2 is constant $\therefore i_{R_2}$ remains unchanged
 Potential difference across $L = 0 \Rightarrow \frac{L di_L}{dt} = 0$
 $\Rightarrow i_L = \text{constant}$
 $i_{\text{battery}} = i_{R_1} + i_{R_2} \Rightarrow \text{Increases}$

5.(ABCD)

(A) $\mu = \frac{1}{\sin C}$

For T.I.R of ray (1)

$30^\circ > C$

$\Rightarrow \sin 30^\circ > \sin C \Rightarrow \frac{1}{2} > \frac{1}{\mu}$

$\Rightarrow \mu > 2 \therefore \mu(\min) > 2$

(B) Snell's law for Ray (2),

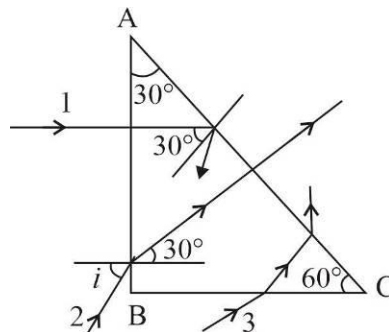
$1 \sin i = 2 \sin 30^\circ$

$\Rightarrow \sin i = 1 \therefore i = 90^\circ$

(C) Deviation of Ray (2) $= i - r = 90^\circ - 30^\circ = 60^\circ$

(D) For Ray(3), $i = e \Rightarrow \delta = 2i - \angle C = 2i - 60^\circ$

$r_1 = r_2 = \frac{A}{2} = 30^\circ \Rightarrow i = 90^\circ \therefore \delta = 180^\circ - 60^\circ = 120^\circ$



6.(ABD)

Switch at position 1

$\tan \frac{\pi}{6} = \frac{\omega L}{R} \Rightarrow \omega L = \frac{R}{\sqrt{3}}$

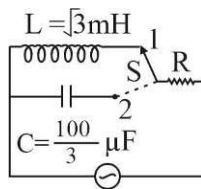
Switch at position 2

$\tan \frac{\pi}{4} = \frac{1}{\omega RC} \Rightarrow \omega RC = 1$

$\therefore \frac{1}{\omega C} = \sqrt{3} \omega L$

$\omega = \frac{1}{\sqrt{\sqrt{3}LC}} = \frac{1}{\sqrt{3 \times 10^{-3} \times \frac{100}{3} \times 10^{-6}}}$

$\therefore f = \frac{\sqrt{2.5}}{\pi} \text{ kHz}; \quad z_1 = \sqrt{R^2 + x_L^2} = \frac{2R}{\sqrt{3}}, \quad z_2 \sqrt{R^2 + x_C^2} = \sqrt{2}R$



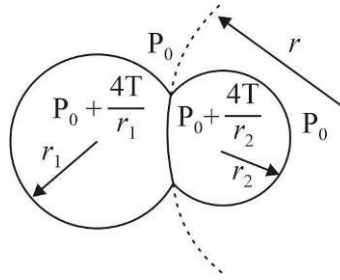
7.(BC)

8.(6) For interface, excess pressure = $\frac{4T}{r}$

$$\Rightarrow \left(P_0 + \frac{4T}{r_2}\right) - \left(P_0 + \frac{4T}{r_1}\right) = \frac{4T}{r}$$

$$\Rightarrow \frac{1}{r_2} - \frac{1}{r_1} = \frac{1}{r} \Rightarrow r = \frac{r_1 r_2}{r_1 - r_2}$$

$$\therefore r = \frac{(3)(2)}{3-2} \text{ cm} = 6 \text{ cm}$$



9.(3) A $\longrightarrow$ B

t = 0	N_0	N_0
t = t	N	$2N_0 - N$

$$\frac{N}{2N_0 - N} = \frac{1}{7} \Rightarrow N = \frac{N_0}{4}$$

2 half lives : 1500 years

10.(144) Let the mass of the ring be m , let its radius be R , and let the angular velocity of the system when the ring is at its lowest position be ω

Let the force exerted by any one rod be T

Due to symmetry at the instant when the rod is at the lowest position, the force exerted by the other rod is also T

Since the centre of mass of the ring is moving in circular path of radius $\frac{2R}{\pi}$,

$$2T \cos 30^\circ - mg = m\omega^2 \left(\frac{2R}{\pi}\right); \quad T = 144 \text{ N}$$

11.(2) Initially, the boy looks at the source directly

After the introduction of the lens, the boy no longer looks at the source itself, but looks at the image of the source formed by the lens

For the position of the image:

$$\frac{1}{v} = \frac{1}{f} + \frac{1}{u} = \frac{1}{60} + \frac{1}{-180}$$

$$\Rightarrow v = 90 \text{ cm}$$

So, the image is at a distance $180 - 90 = 90 \text{ cm}$ from the boy

Now, the image itself is M times the size of the source of light, where

$$M = \left|\frac{v}{u}\right| = \frac{1}{2}$$

Since the perceived size is proportional to $\frac{1}{r}$, the final size of the image as perceived by the boy is

$$\left(\frac{\frac{1}{90}}{\frac{1}{360}}\right) \left(\frac{1}{2}\right) = 2 \text{ times the size initially perceived}$$

12.(4) Speed of sound in the wire, $c = \sqrt{\frac{50}{0.02}} = 50 \text{ m/s}$

So, the beat frequency, $f_B = \frac{50}{2(0.25)} - \frac{50}{2(0.26)} \approx 4$

13.(B) For constant velocity

$$q[\vec{E} + (\vec{V} \times \vec{B})] = \vec{0} \Rightarrow \vec{E} = \vec{B} \times \vec{V}$$

14.(B) For helical path, having constant pitch in absence of $\vec{E}$, $\vec{B} \cdot \vec{V} \neq 0$

$$\vec{B} \times \vec{V} \neq \vec{0}$$

15.(B) For cycloid path,

$\vec{E}$ and $\vec{B}$ must be uniform and constant mutually perpendicular fields ($\vec{E}$ as well $\vec{B}$). Also, initial velocity must be zero.

16.(C) The process must be isochoric cooling.

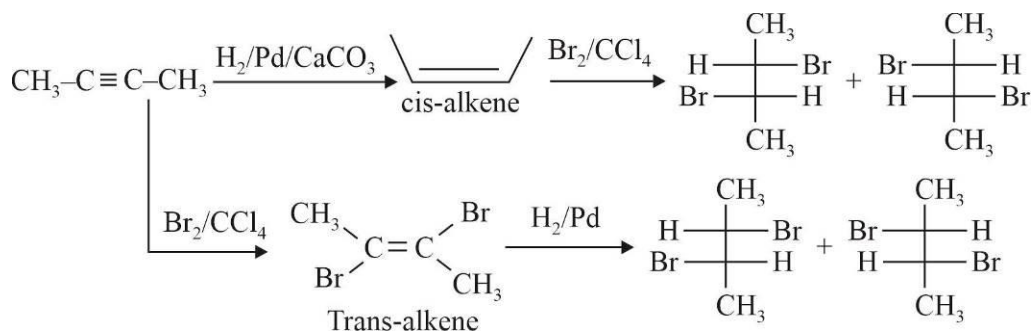
17.(B) The correct combination is for isobaric process

18.(C) The process must be isothermal which was used by Newton for sound speed in air

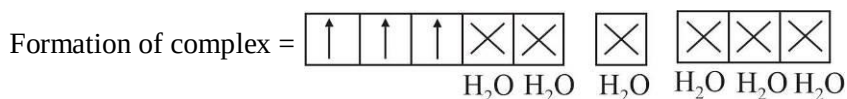
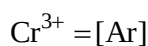
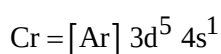
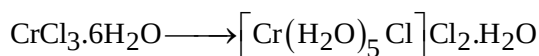
CHEMISTRY

1.(AC)

2.(AB)



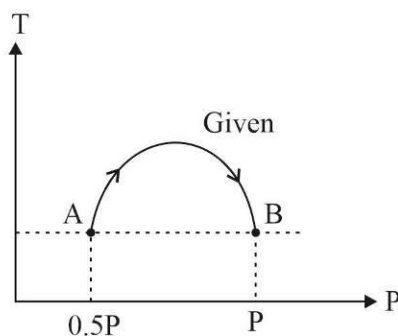
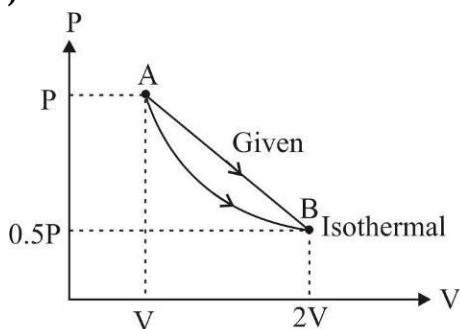
3.(ABC)

Hybridisation $\Rightarrow d^2sp^3$ Magnetism $\Rightarrow$ paramagneticIt will exist in hydrated isomerism $[\text{Cr}(\text{H}_2\text{O})_5\text{Cl}]\text{Cl}_2 \cdot \text{H}_2\text{O}, [\text{Cr}(\text{H}_2\text{O})_6]\text{Cl}_3$.

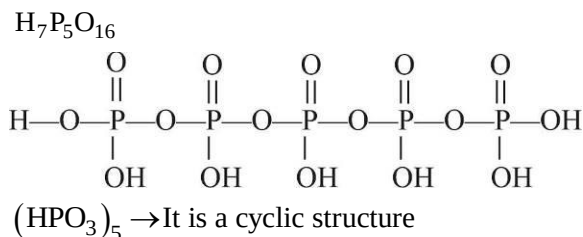
4.(BD)

At point Z, mole fractions of X and mole fractions of Y in vapour phase is 0.5. From the graph, $P_X^0 > P_Y^0$ so intermolecular force of interaction of X – X in pure liquid is less than Y – Y. $P_T = 240 + 240 = 480 \text{ mm}$, at point Z.

5.(ABD)



6.(AD)



7.(AC)

e^- transition from HOMO-LUMO in visible region and presence of unpaired electron gives it a blue colour.

8.(3) $\text{N}_2^+ = 13e^-$ $\text{B.O} = \frac{9-4}{2} = 2.5$ (It has one unpaired e^- in σ_{2p_z})

$\text{O}_2^- = 17e^-$ $\text{B.O} = \frac{10-7}{2} = 1.5$

It has one unpaired e^- in π_{2p_y}

$\text{OF} = 17e^-$ $\text{B.O} = \frac{10-7}{2} = 1.5$

It has one unpaired e^- in π_{2p_y}

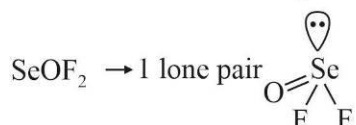
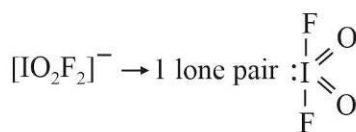
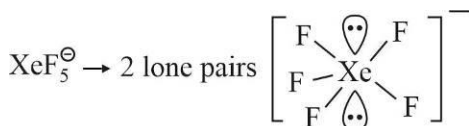
9.(4) Anti-aromatic species follow $(4n)\pi e^-$ rule : (1, 4, 6, 7) are antiaromatic species.

10.(4) $\lambda_m = \frac{k \times 1000}{M}$

$200 = \frac{1}{R} \times \frac{\ell}{A} \times \frac{1000}{M} \Rightarrow 200 = \frac{I}{V} \times \frac{\ell}{A} \times \frac{1000}{M}$

$200 = \frac{I}{10} \times \frac{8}{4} \times \frac{1000}{0.04} \Rightarrow I = \frac{40}{1000} = 0.04 \text{ A} = 40 \text{ mA}$

11.(5)



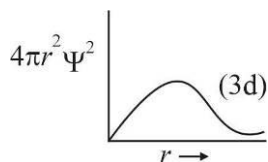
Sum of lone pair of electrons = $2 + 1 + 1 + 1 = 5$

12.(2) Volume of metal taken $= \frac{m}{d} = \frac{100}{6.25} = 16 \text{ cm}^3$

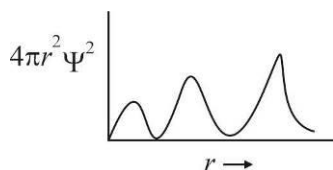
Volume of each unit cell $= (2 \times 10^{-8} \text{ cm})^3 = 8 \times 10^{-24} \text{ cm}^3$

No. of unit cells $= \frac{16}{8 \times 10^{-24}} = 2 \times 10^{24}$

13.(C) 3d subshell – two nodal plane (ℓ)

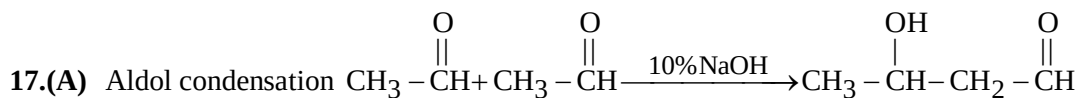
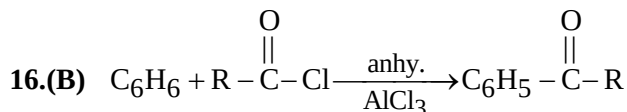
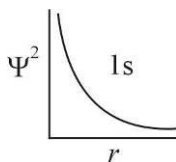


14.(B) 3s subshell contain 3 maxima $[(n - \ell) = (3 - 0) = 3]$ and 2 radial nodes

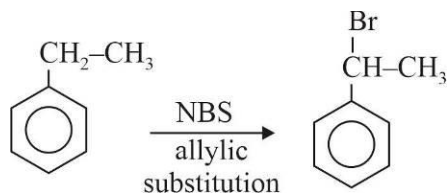


15.(A) 1s orbital $\rightarrow$

And $\Psi_{100} \propto \left(\frac{z}{a_0}\right)^{3/2} e^{-\left(\frac{zr}{a_0}\right)}$



18.(C)



Follows free radical mechanism.

MATHEMATICS

1.(ABCD)

We have $P(A \cup B) \geq \max \{P(A), P(B)\} = \frac{2}{3}$,

Next, $P(A \cap B) = P(A) + P(B) - P(A \cup B) \geq P(A) + P(B) - 1 = 1/6$

And $P(A \cap B) \leq \min \{P(A), P(B)\} = \frac{1}{2} \Rightarrow \frac{1}{6} \leq P(A \cap B) \leq \frac{1}{2}$

Also $P(A \cap B^c) = P(A) - P(A \cap B) \leq \frac{1}{2} - \frac{1}{6} = \frac{1}{3}$

Lastly $P(A' \cap B) = P(B) - P(A \cap B)$

$3/2 - 1/2 \leq P(A' \cap B) \leq \frac{1}{3} - \frac{1}{6} \Rightarrow 1/6 \leq P(A' \cap B) \leq 1/2$

2.(BC)

$T \equiv Xdy - Ydx = xdy - ydx$ and $N = Xdx + Ydy = xdx + ydy$ $|xdy - ydx| = |xdx + ydy|$

$$\Rightarrow \frac{1}{2} d(x^2 + y^2) = \pm (xdy - ydx)$$

$$\Rightarrow \frac{d(x^2 + y^2)}{2(x^2 + y^2)} = \pm \left(\frac{xdy - ydx}{x^2 + y^2} \right)$$

$$\Rightarrow \ln \sqrt{x^2 + y^2} = \pm \tan^{-1} \left(\frac{y}{x} \right) + c \Rightarrow \sqrt{x^2 + y^2} = ce^{\pm \tan^{-1} \left(\frac{y}{x} \right)}$$

3.(ABCD)

A, D is true as PQR is equilateral triangle so orthocentre, circumcentre and centroid will coincide.

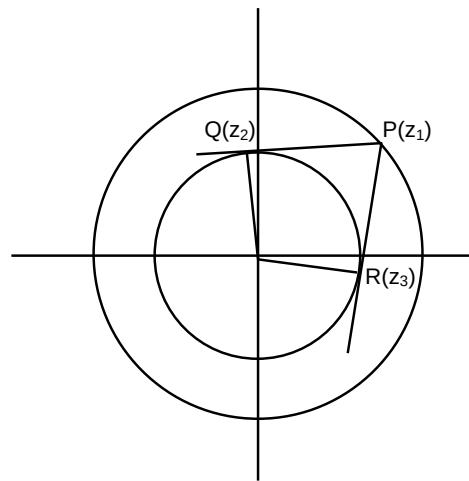
$$(B) \left| \frac{z_1 + z_2 + z_3}{3} \right| = 1$$

$$\Rightarrow |z_1 + z_2 + z_3|^2 = 9$$

$$(z_1 + z_2 + z_3)(\bar{z}_1 + \bar{z}_2 + \bar{z}_3) = 9$$

$$\left(\frac{4}{\bar{z}_1} + \frac{1}{\bar{z}_2} + \frac{1}{\bar{z}_3} \right) \left(\frac{4}{z_1} + \frac{1}{z_2} + \frac{1}{z_3} \right) = 9.$$

$$(C) \angle QOR = 120^\circ.$$



4.(ABCD)

We know $y = mx \pm \sqrt{a^2 m^2 - b^2}$ is tangent to (1) for all m.

Similarly $y = m_1 x \pm \sqrt{(-b^2)m_1^2 - (-a^2)}$.

i.e. $y = m_1 x \pm \sqrt{a^2 - b^2 m_1^2}$ is tangent to (2)

for common tangent

$$m = m_1 \text{ and } a^2 m^2 - b^2 = a^2 - b^2 m_1^2$$

$$\text{i.e. } (a^2 + b^2)(m^2 - 1) = 0 \Rightarrow m = \pm 1$$

$$\therefore \text{ Equation of common tangent is } y = \pm x \pm \sqrt{a^2 - b^2}.$$

5.(ABCD)

$$y' = \frac{1}{\sqrt{1 - \left(\frac{2x}{1+x^2}\right)^2}} \cdot \frac{2(1+x^2) - 4x^2}{(1+x^2)^2} = \frac{2(1-x^2)}{\sqrt{(1-x^2)^2(1+x^2)}}$$

$$y' = \begin{cases} \frac{2}{1+x^2} & \text{for } |x| < 1 \\ -\frac{2}{1+x^2} & \text{for } |x| > 1 \end{cases}$$

Hence, for $|x| = 1$, the derivative doesn't exist.

6.(A) For non-diagonal entries, we required even no. of 1, even number of -1 and even no. of 0, for diagonal three entries remain, $-1, 0, 1$. So no. of cases in which trace = 0 are $3!$ And no. of symmetric matrices for each arrangement of 1, $-1, 0$ in diagonal = $3!$

$$\text{Total such matrices} = 3! \times 3! = 36$$

7.(CD)

We know semi-latus rectum is harmonic mean of segments of focal chord

i.e. $\cos(\alpha - \beta), \cos \alpha, \cos(\alpha + \beta)$ are in H.P

$$\Rightarrow \cos \alpha = \frac{2 \cos(\alpha + \beta) \cos(\alpha - \beta)}{\cos(\alpha + \beta) + \cos(\alpha - \beta)} \Rightarrow \cos \alpha = \frac{\cos 2\alpha + \cos 2\beta}{2 \cos \alpha \cos \beta} \Rightarrow 2 \cos^2 \alpha \cos \beta = 2 \cos^2 \alpha + 2 \cos^2 \beta - 2$$

$$\cos^2 \alpha (\cos \beta - 1) = \cos^2 \beta - 1$$

$$\cos \beta = 1 \text{ or } \cos^2 \alpha \sec^2 \frac{\beta}{2} = 2 \Rightarrow \cos \alpha \sec \frac{\beta}{2} = \pm \sqrt{2}.$$

$$8.(2) \quad a_{ij} = 0 \quad \forall i \neq j \text{ and } a_{ij} = (n-1)^2 + i \quad \forall i = j$$

$$\text{Sum of all the elements of } = (2n-1)(n-1)^2 + (2n-1)n = 2n^3 - 3n^2 + 3n - 1 = n^3 + (n-1)^3$$

$$\text{So, } T_n = (-1)^n [n^3 + (n-1)^3] = (-1)^n n^3 - (-1)^{n-1} (n-1)^3 = V_n - V_{n-1}$$

$$\Rightarrow \sum_{n=1}^{102} T_n = \sum_{n=1}^{102} (V_n - V_{n-1}) = V_{102} - V_0 = (102)^3$$

$$\left[\frac{\sum_{n=1}^{102} T_n}{520200} \right] = 2$$

$$9.(7) \quad \text{Given sum} = \{1^3 + 2^3 + 3^3 + \dots + n^3\} - 2 \times 2^3 \left\{ 1^3 + 2^3 + 3^3 + \dots + \left(\frac{n-1}{2}\right)^3 \right\}$$

$$10.(8) \quad 12 \lim_{t \rightarrow 0} \frac{A(t)}{B(t)} = \lim_{t \rightarrow 0} \frac{12 \int_0^t \sin^2 x \, dx}{\frac{1}{2} t \cdot \sin t^2} \Rightarrow \lim_{t \rightarrow 0} \frac{24 \sin t^2}{\sin t^2 + 2t^2 \cos t^2} = \frac{24}{3} = 8$$

11.(1601)

Given lines are

$$(2 + c)x + 5c^2y = 1 \quad \dots (1)$$

$$3x + 5y = 1 \quad \dots (2)$$

Solving (1) and (2), we get

$$x = \frac{c^2 - 1}{3c^2 - c - 2} \text{ and } y = \frac{c - 1}{-5(3c^2 - c - 2)}$$

point of intersection when $c \rightarrow 1$, will be given by

$$\left(\lim_{c \rightarrow 1} \frac{(c-1)(c+1)}{(c-1)(3c+2)}, \lim_{c \rightarrow 1} \frac{c-1}{-5(c-1)(3c+2)} \right) \text{ i.e. } \left(\frac{2}{5}, -\frac{1}{25} \right) \text{ is the center of the circle}$$

Therefore, the equation of the required circle is

$$\left(x - \frac{2}{5} \right)^2 + \left(y + \frac{1}{25} \right)^2 = \left(2 - \frac{2}{5} \right)^2 + \left(\frac{1}{25} \right)^2 \text{ i.e. } (25x - 10)^2 + (25y + 1)^2 = 1601$$

12.(1650)

$$N = 3^9 - {}^3C_1 \cdot 2^9 + {}^3C_2 = 3[3^8 - 2^9 + 1] = 3[6561 - 511] = 3 \times 6050 = 3 \times 550 \times 11$$

13.(B) 14.(A) 15.(C)

A. $x^2 = 4y$, let midpoint of chord is (h, k) $T = S_1hx - 2(k + y) = h^2 - 4k$

$$\Rightarrow hx - 2y + 2k - h^2 = 0$$

$$2y = hx + 2k - h^2$$

line is tangent to the Parabola $x^2 + 4y = 0$

$$x^2 + 2a(hx + 2k - h^2) = 0$$

$$D = 0 \Rightarrow (2ah)^2 = 4(1)(2a)(2k - h^2)$$

$$\Rightarrow 4a^2h^2 = 4 \times 2a(2k - h^2) \Rightarrow ax^2 = 4y - 2x^2 \Rightarrow x^2(a + 2) = 4y$$

B. $x^2 - y^2 = b^2$

$$T = S_1 \Rightarrow hx - ky - b^2 = h^2 - k^2 - b^2$$

$$\text{Solve : } \frac{h}{k} = a \quad h = ak$$

$$\text{Locus } x = ay$$

C. Chord of contact is normal

$$T = 0$$

$$Ky - 2a(x + h) = 0 \text{ is normal to Parabola } y^2 = 4ax$$

$$Ky = 2ax + 2ah$$

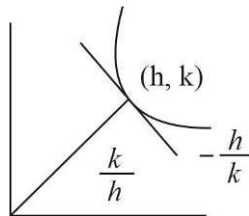
$$y = \frac{2ax}{k} + \frac{2ah}{k}$$

$$m = \frac{2a}{k}$$

$$c = -2am - am^3$$

$$\begin{aligned}\frac{2ah}{k} &= 2a \frac{2a}{k} - a \left(\frac{2a}{k} \right)^3 \\ \Rightarrow h &= -2a - a \left(\frac{2a}{k} \right)^2 \\ h + 2a + \frac{4a^3}{k^2} &= 0 \\ y^2(x + 2a) + 4a &\end{aligned}$$

D.



Equation of Tangent is

$$y - k = \frac{-h}{k}(x - h)$$

$$hx + ky = h^2 + k^2$$

$ky - k^2 = -hx + h^2$ is tangent to curve $xy = a^2$

$$xy = a^2$$

$$\left(at, \frac{a}{t} \right)$$

$$x \frac{a}{t} + y at = 2a^2$$

$$\frac{x}{t} + ty = 2a$$

$$\frac{h}{\frac{1}{t}} = \frac{k}{t} = \frac{h^2 + k^2}{za}$$

$$t = \frac{h^2 + k^2}{zah} \quad \frac{1}{t} = \frac{h^2 + k^2}{zak}$$

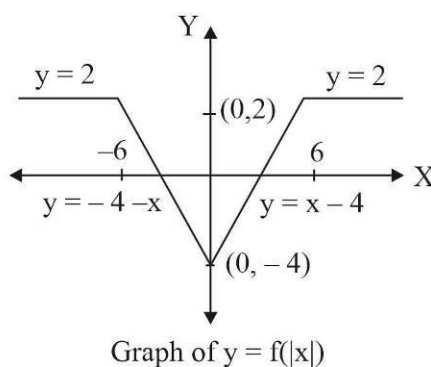
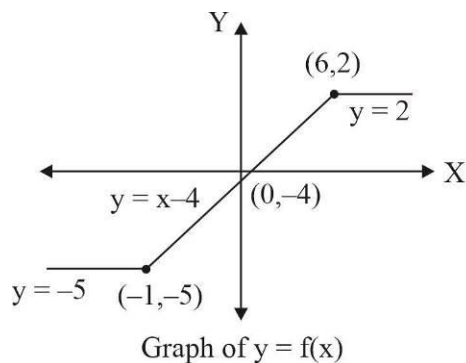
$$1 = \frac{(h^2 + k^2)^2}{4a^2hk}$$

$$(x^2 + y^2)^2 = 4a^2xy$$

16.(C) 17.(D) 18.(A)

A. Clearly, $|x^2 - 9| + |x^2 - 4| = 5 = |(x^2 - 9) - (x^2 - 4)|$
 So, $(x^2 - 9)(x^2 - 4) \leq 0$ (As $|a| + |b| = |a - b| \Rightarrow ab \leq 0$)
 $\Rightarrow x \in [-3, -2] \cup [2, 3]$

B.



Clearly, $R_{f(|x|)} = [-4, 2]$

$\therefore$ Number of integers in the range of function $f(|x|) = 7$

$$\text{Now, clearly } \lim_{x \rightarrow n} \frac{[x]^2 - 13[x] + 42}{(x-7)(x-6)} = \lim_{x \rightarrow 7} \frac{([x]-7)([x]-6)}{(x-7)(x-6)} = 0$$

C.

$$L = \lim_{x \rightarrow 0} \frac{a \left(x - \frac{x^3}{3!} + \dots \right) + bx \left(1 + \frac{x}{1!} + \frac{x^2}{2!} + \dots \right) + 3x^2}{\left(\frac{\sin x - x}{x^3} - \frac{(x - \tan x)}{x^3} \right) x^3}$$

$$L = 6 \lim_{x \rightarrow 0} \frac{x(a+b) + x^2(b+3) + x^3 \left(\frac{b}{2} - \frac{a}{6} \right) + \dots}{x^3} \Rightarrow L = 6 \left(\frac{b}{2} - \frac{a}{6} \right)$$

$$\Rightarrow a + b = 0 \text{ and } b + 3 = 0 \Rightarrow b = -3. \text{ Also } a = 3$$

$$L = 6 \left(-\frac{3}{2} - \frac{3}{6} \right) = -12$$

D.

$$\text{We have } 1 - \frac{1}{{}^{m+1}C_2} = 1 - \frac{2}{(m+1)m} = \frac{m^2 + m - 2}{m(m+1)} = \frac{(m+2)(m-1)}{(m+1)m}$$

$$\text{Hence, } P_n = \prod_{m=2}^n \frac{m-1}{m} \cdot \prod_{m=2}^n \frac{m+2}{m+1} = \left(\frac{1}{2} \times \frac{2}{3} \times \frac{3}{4} \times \dots \times \frac{n-1}{n} \right) \left(\frac{4}{3} \times \frac{5}{4} \times \frac{6}{5} \times \dots \times \frac{(n+2)}{(n+1)} \right)$$

$$\text{Hence, } P_n = \frac{1}{n} \left(\frac{n+2}{3} \right) = \frac{n+2}{3n}$$

$$\lim_{n \rightarrow \infty} P_n = \frac{1}{3}$$